# Topics For Jeopardy-style or Attack-Defense CTF

## Web Exploitation

- SQL Injection
- Cross-Site Scripting (XSS)
- Cross-Site Request Forgery (CSRF)
- Directory Traversal
- Local File Inclusion (LFI) and Remote File Inclusion (RFI)
- Command Injection
- Server-Side Request Forgery (SSRF)

## Cryptography

- Symmetric and Asymmetric Encryption
- Hashing and Cracking Hashes (MD5, SHA-1, etc.)
- RSA and Public Key Exploitation
- XOR-based Ciphers
- Steganography
- Analyzing Encodings (Base64, Hex, URL, etc.)

## Reverse Engineering

- Disassembling Binary Files
- Static and Dynamic Analysis Tools (Ghidra, IDA Pro, Radare2)
- Understanding Assembly Code
- Debugging with gdb
- De-obfuscation Techniques
- Patch Binary Exploits

## Forensics

- Memory Forensics
- Disk Image Analysis

Network Packet Analysis (using Wireshark or tcpdump)

Metadata Analysis

Log Analysis

File Recovery

## Binary Exploitation

Buffer Overflow

Return-Oriented Programming (ROP)

Format String Vulnerabilities

Heap Exploitation

Exploiting Modern Protections (ASLR, NX, PIE, etc.)

## Networking

Packet Capture Analysis

Exploiting Network Misconfigurations

Understanding Protocols (HTTP, DNS, FTP, etc.)

Analyzing ARP Spoofing or Man-in-the-Middle Attacks

## OSINT (Open Source Intelligence)

Searching for Clues Using Public Information

Email and Social Media Analysis

Reconnaissance with Tools (e.g., Shodan, Maltego)

## Miscellaneous

Python Scripting for Automation

Regular Expressions for Pattern Matching

File Format Analysis (PDF, Images, ZIP, etc.)

Using Exploit Frameworks like Metasploit

Capture Traffic in CTFs (e.g., PCAP challenges)

## Tools to Introduce

Web: Burp Suite, OWASP ZAP

Crypto: CyberChef, hashcat

Forensics: Autopsy, Volatility

Binary: pwndbg, peda, angr

OSINT: The Harvester, Recon-ng

Networking: Wireshark, Scapy